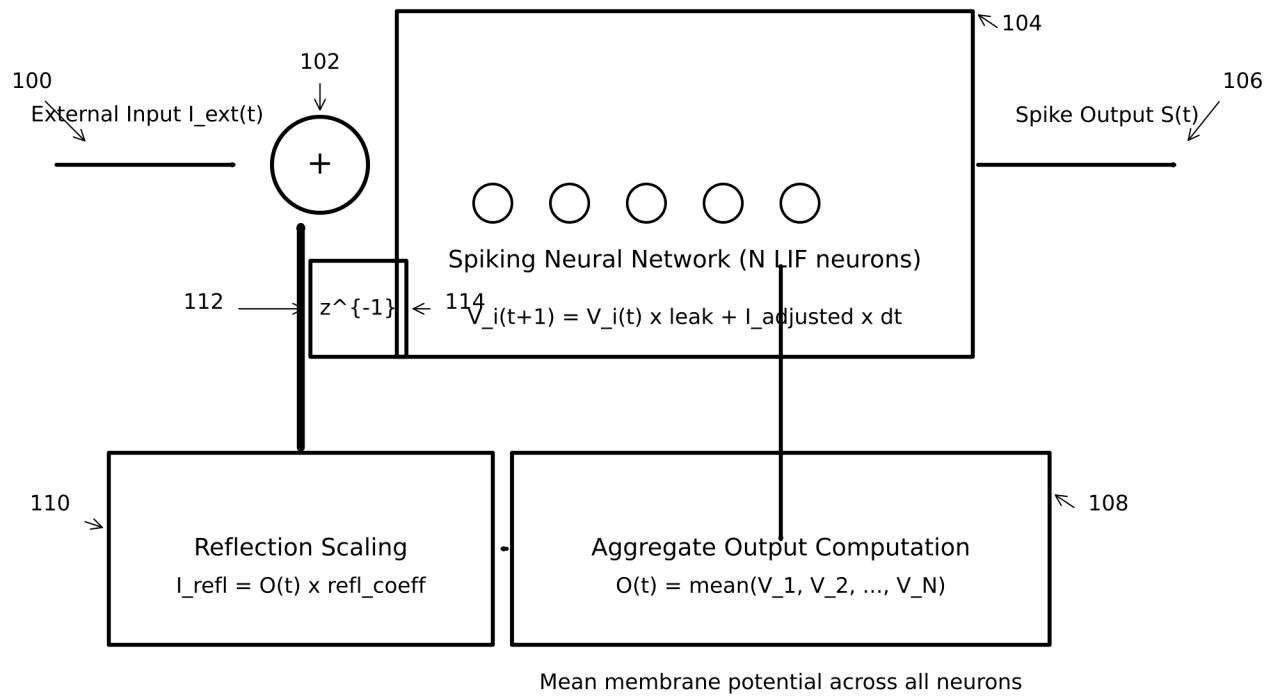




Self-observation: the network observes its own prior aggregate state

$$I_{\text{adjusted}}(t+1) = I_{\text{ext}}(t+1) + I_{\text{reflection}}(t)$$

FIG. 1

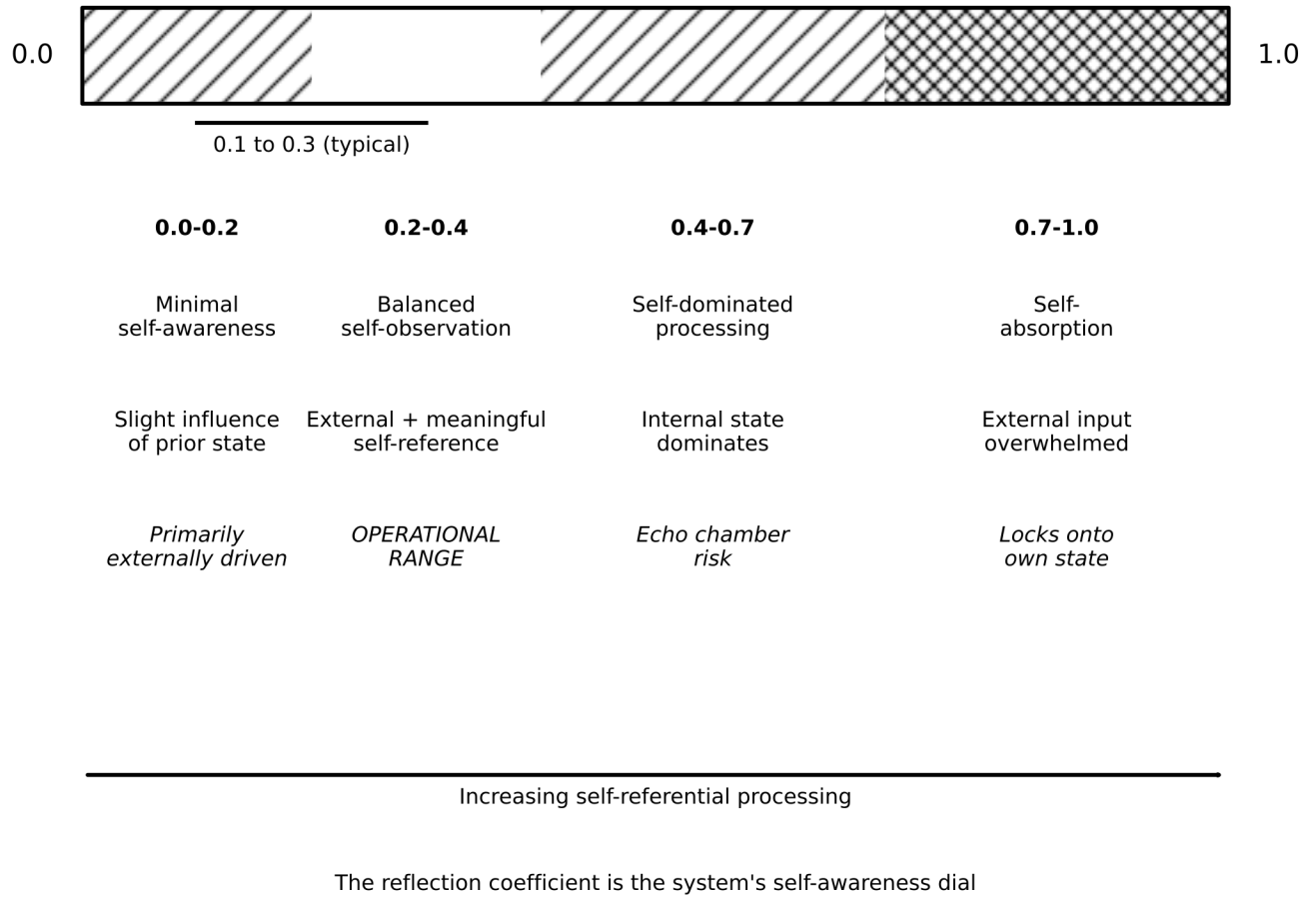


FIG. 2

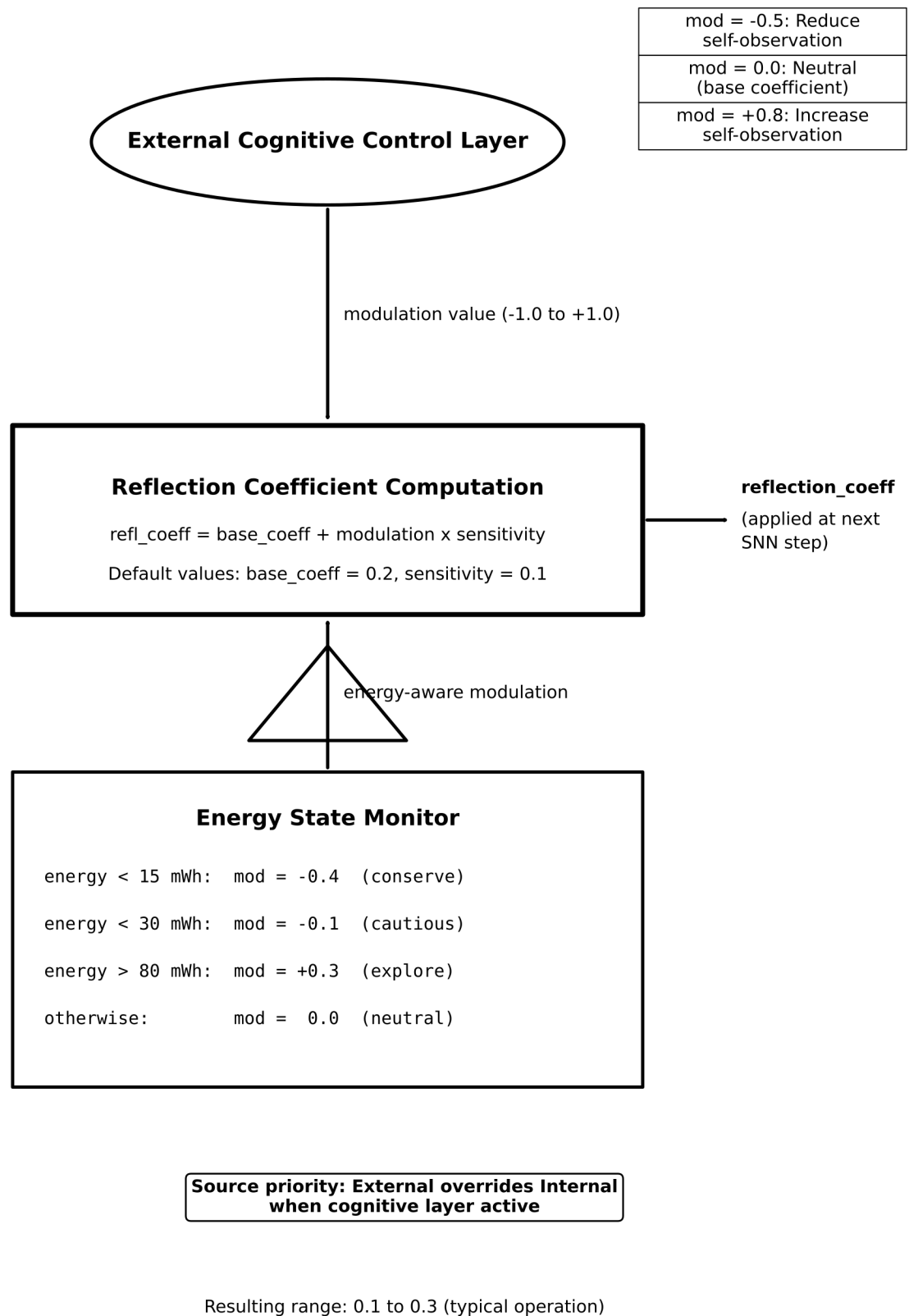


FIG. 3

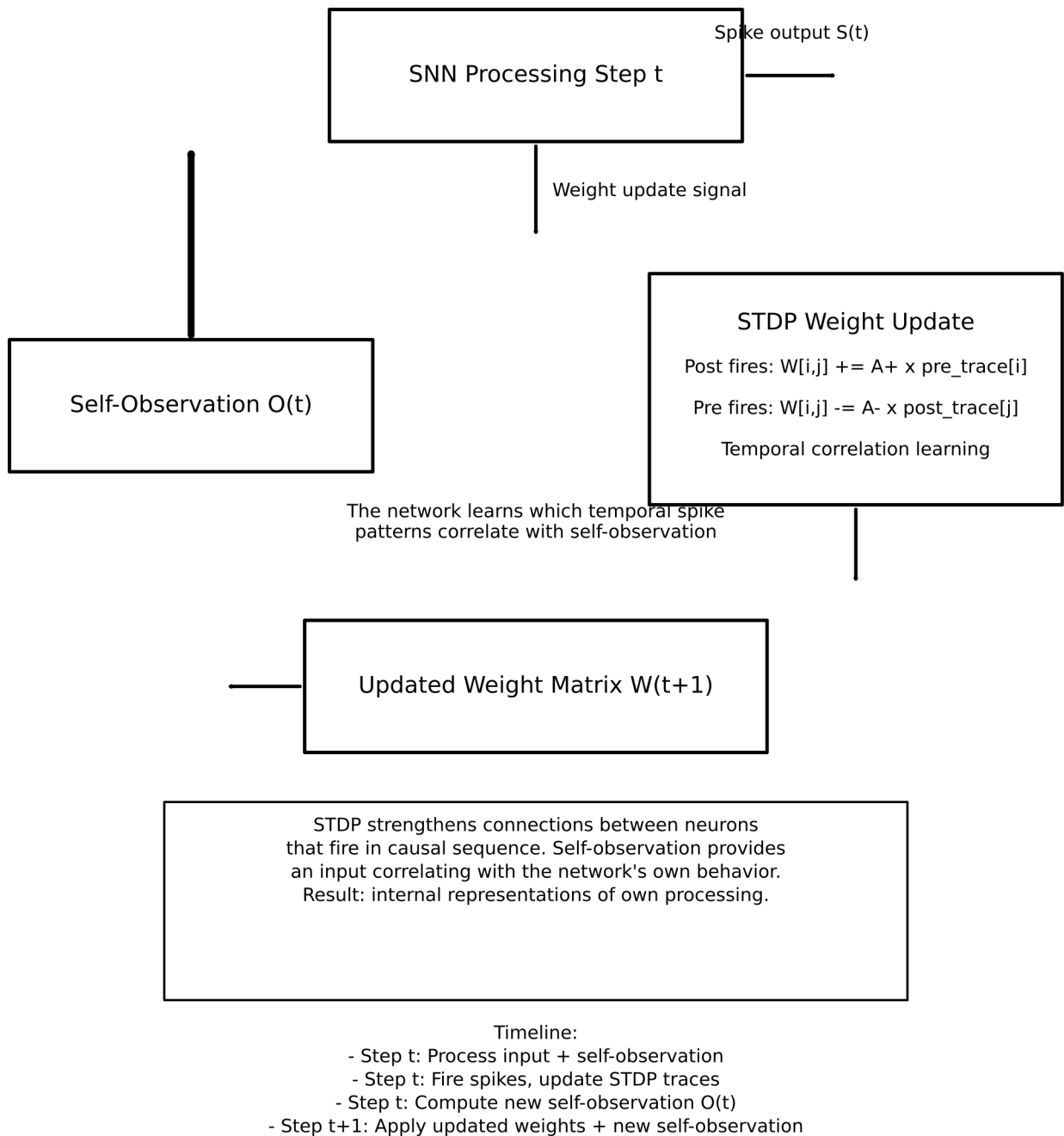
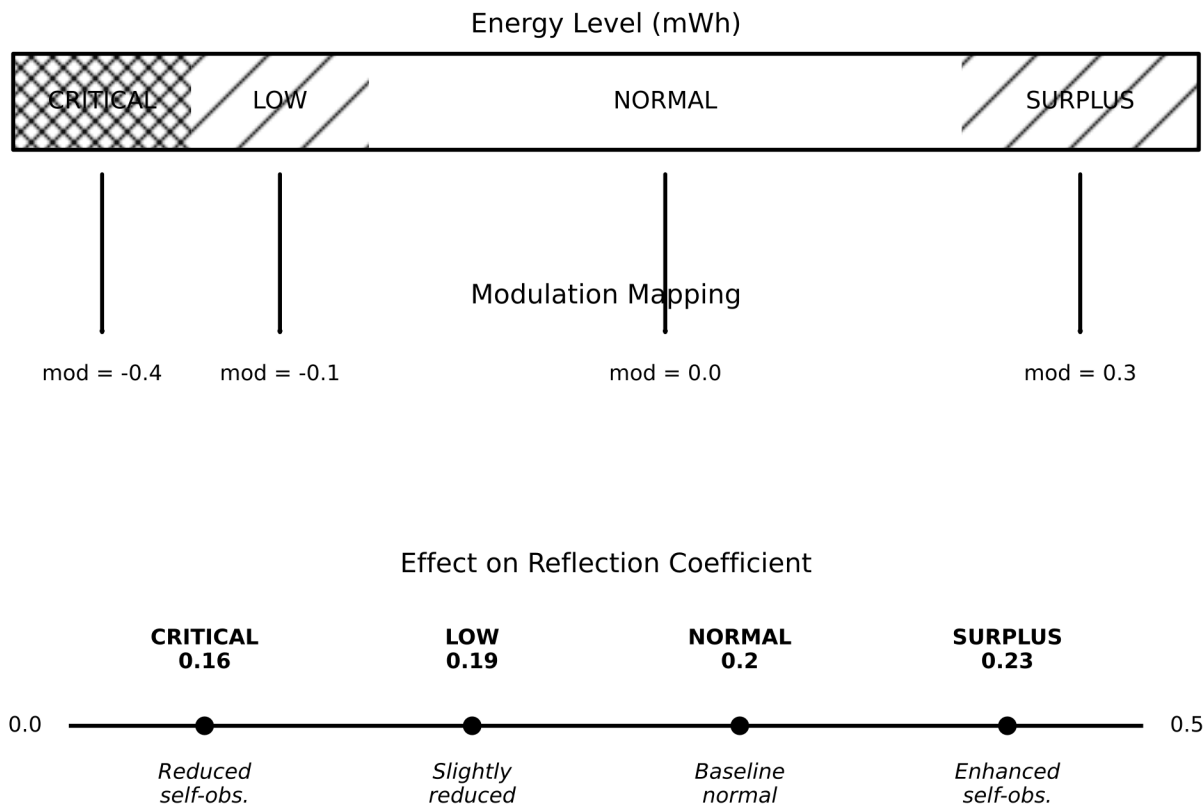


FIG. 4



Energy-aware regulation ensures the system reduces cognitive overhead when energy is scarce and explores deeper self-reference when energy is abundant

FIG. 5

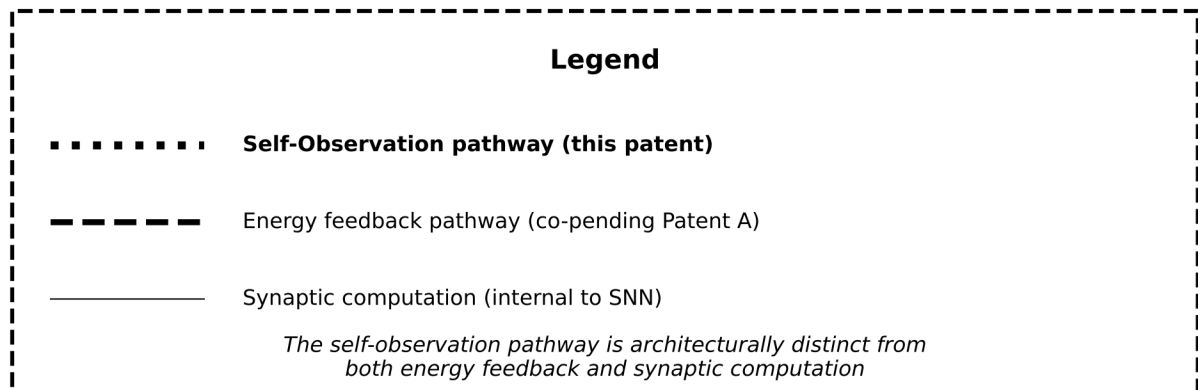
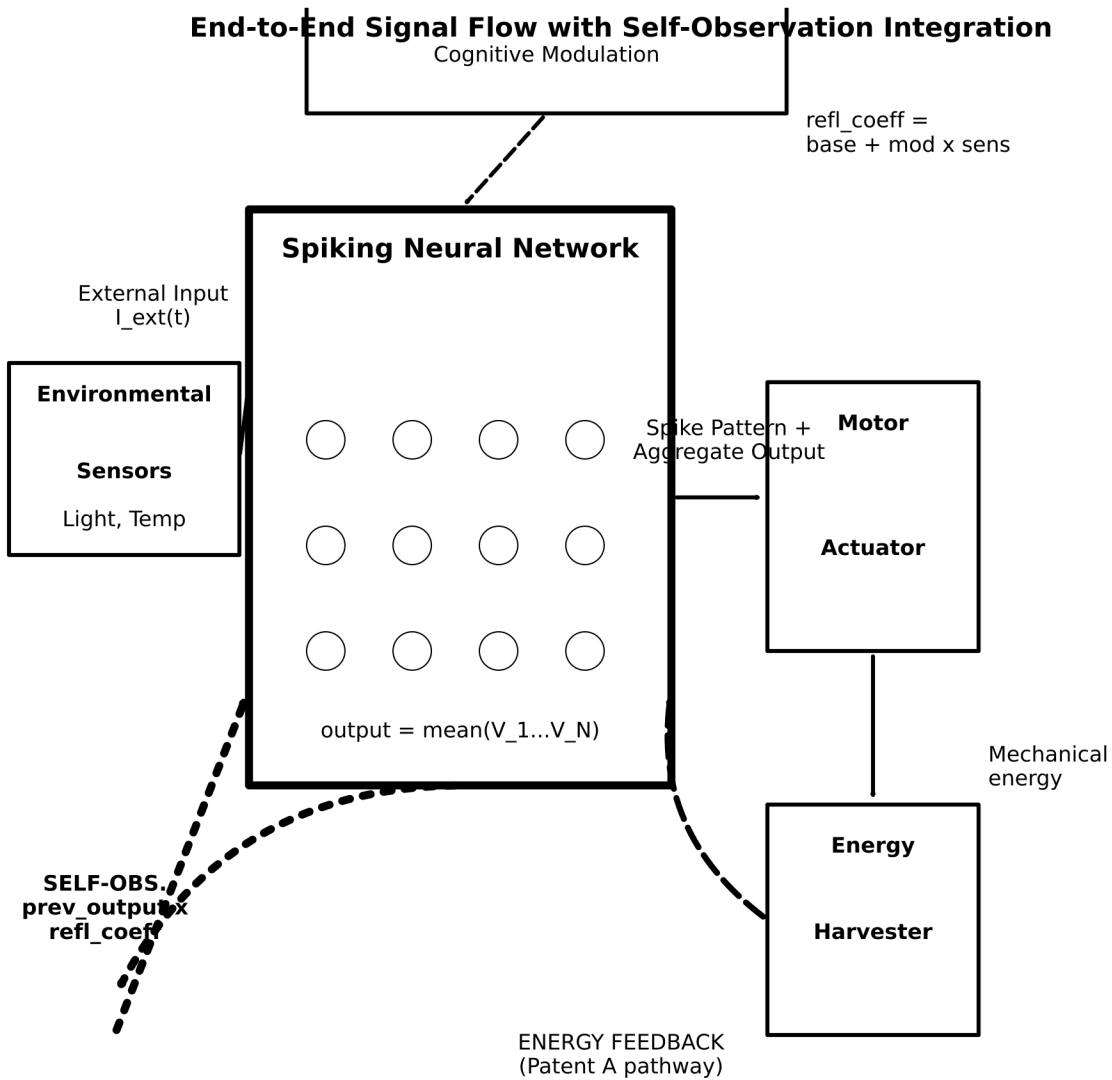


FIG. 6